

Generic BCI Classifiers for Discrimination of Motor Imagery of Left/Right Steps and Forward Walking

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Abstract—Brain-computer interfaces (BCI) have been used to control the gait of a virtual self-avatar, using motor imagery of the feet, in order to restore motor control in gait rehabilitation. The considerable training time required to use such a BCI is an obstacle to their adoption in a clinical setting. One technique used to enhance BCI control and to shorten training time is to eliminate offline calibration using a generic classifier that is pre-trained over many participants, each performing many trials. This paper investigates the performance of generic models that were derived from 2 datasets, each containing the data of 20 participants. They participated in a sequential training to control the gait of an avatar when cued to imagine a single step forward using their left or right foot, or to start walking forward. The avatar moved in response to two calibrated RLDA classifiers that used the μ PSD over the foot area of the motor cortex as features. The generic models were tested on the offline and online data of the participants. The models performed as well as models obtained from participant-specific offline data with a mean performance of 86%. The results show the possibility of designing a participant-independent, zero-training lower-limb MI-BCI.

Keywords—Brain-computer interface, Virtual reality, EEG, Classification, Avatar, Gait, Generic classifier.

I. INTRODUCTION

A BCI is a system that measures brain activity of an intention to do something and converts it into a control command that replaces, restores, enhances, supplements or improves natural brain activity output [1]. In the field of rehabilitation, this control command has been used to control external devices such as robotic and prosthetic devices [2] and a lower limbs exoskeleton [3] as well as to control the gait of a virtual self-avatar [4]. When such systems are controlled through motor imagery (MI) of the desired actions (e.g. individual left and right steps as well as walking forward), this can trigger neural circuitry reorganization [5] and restore motor control during gait rehabilitation [6].

When designing a BCI, the most important step is tuning the classifier [7]. When fed with different MI patterns produced by the user, the classifier “learns” to discriminate these distinct patterns and translate the extracted signal features into control commands. For an effective BCI design and control performance, the right classifier and tuning parameters must be chosen very carefully [7]. Powerful algorithms have been used

to counter-act the problems and limitations of some of classification algorithms, such as using regularization parameters to counter-act overfitting [8]. Since the classifier needs to be tuned for a specific user, a large amount of data from each user must be fed to the classifier. To counter this limitation, many studies have suggested to eliminate offline calibration using a generic classifier that is pre-trained over many participants, using many trials [9]. This generic classifier is used to train participants to control a participant-optimized BCI [10].

For example, Lotte et al. [11] proposed to use a participant-independent P300 BCI, previously learnt from the data of many other participants. Their BCI resulted in a better performance, and a shorter training time, reduced from 40 minutes to 2 minutes. Similar results were found by other studies by running a generic LDA classifier in a MI-BCI [12]. Their findings state that, even though the inter-participant variability poses a challenge, all participants scored high performance [12]. Generic classifiers based on the data of 80 participants have also been used by Vidaurre et al. [13], who showed that their performance is significantly better (an increase of 13%) than the state-of-the-art classical classification approach. However, these previous studies used generic classifiers for an upper-limb MI-BCI. To our knowledge, there are no generic classifiers that have been used for lower-limb MI-BCIs.

The objective of this study was to investigate the possibility of using generic classifiers to shorten the time required to learn to operate a BCI that controls the gait of an avatar and to enhance the performance of this BCI. Different sparse-data algorithms were investigated in order to select the algorithm that would yield the highest classification accuracy. The training and test datasets were derived from the two previous studies [14, 15]. The generic models were tested only offline, on different tests.

II. MATERIALS AND METHODS

A. Experiment Design and Setup

The datasets used for this study were derived from two previous studies that recruited 20 participants each. During both experiments, participants were looking at the lower limbs of their self-avatar through a head-mounted display (HMD). The VE consisted of a virtual hallway with horizontal lines on the

floor in order to for forward movement to be more easily perceived while looking at one's feet.

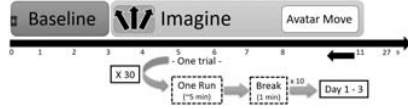


Figure 1: Experimental design of one trial

The VE consisted of a virtual hallway with horizontal lines on the floor in order to for forward movement to be more easily perceived while looking at one's feet. The training of study 1 consisted of a one-day session of 240 trials. The training of study 2 consisted of three sessions on consecutive days with 300 trials on each day (Figure 1). In both studies, each trial lasted approximatively between 7 and 11 s. A trial started with a beep, followed by a three-second pause in order to acquire a baseline EEG recording. An arrow was then presented on the virtual floor in front of the participant for a period of 1.25 s. This arrow either pointed to the left, cueing a step with the left foot; to the right, cueing a step with the right foot; or forward, cueing to imagine walking forward. Upon receiving the cue, participants imagined the appropriate action. On days 2 and 3 of study 2, the avatar was controlled by the participant's brain activity (day 1 was training only). The BCI classifiers were retrained on each day using the data from the previous days. In order to change from the no-movement to the movement state, and to prevent rapid changes, the participant had to accumulate more than a "selection time" (ST) with the correction movement selected [16, 17]. The selection time is the number of successive correct classifier outputs resulting from motor imagery of a movement. ST was set to 3 consecutive outputs, i.e. 600 ms.

B. EEG Data Recording and Pre-Processing

EEG was recorded using the 19-electrode Smart BCI system. These electrodes were grounded to AFz and referenced to both ears using an ear clip on each ear. The electrodes covered the whole scalp with a higher density above the pre-motor, motor and parietal areas. The signals were band-pass filtered (18th order butterworth IIR filter) into 10-13Hz (higher μ band). Artifacts and noise were automatically rejected using MARA, an automatic ICA and noise rejection algorithm [18]. Signals were then detrended with a baseline removal algorithm that used the 200ms preceding the apparition of the arrow.

C. EEG Feature Extraction and Selection

μ PSD of every channel, and 5 PSD asymmetrical ratios (1-sec hanning) were chosen for the feature sets from 20 different 200 ms epochs. To reduce the number of features, the Wilcoxon test and cross-correlation were used to select distinctive features and reduce the dataset by removing features that were highly correlated.

D. Participant-Specific Classification Models

The selected features were normalized [19] and then fed into 2 RLDA classifiers. Classifier 1 (C11) was trained to distinguish between right steps, left steps and no movement. Classifier 2

(C12) was trained to distinguish forward walking from no movement.

1- Regularized Linear Discriminant Analysis (RLDA)

Used in many MI-BCI studies [20], the regularization amount (λ) increases larger eigenvalues of the covariance matrix while decreasing smaller ones, therefore creating a pooled-covariance matrix that is corrected for the bias when estimating sample-based eigenvalues [21].

Thus, regularization improves classification performance by:

- 1) providing generalization and preventing overfit
- 2) decreasing calculation time compared to other classification methods [22].

LDA models were varied by adjusting the regularization amount (λ) which was optimized by setting different values of λ over the range of [0, 1.0] in 0.1 increments [21] in a random search and was validated by a 10-fold cross-validation .

1- Fitting Linear Regression Models (LRM)

The linear regression models were varied by using two choices of penalty hyperparameters: regularization amount (λ) and regularization weight (α), which varied between Lasso (L1), Ridge (L2) and Elastic Net (L3) [23].

E. Generic Classification Models

Final classification models were built by training generic RLDA models and training generic LRMs. Grid search was used for tuning parameters.

F. Protocol for Testing the Generic Classification Models

Six different tests were run over both classifiers (C11 and C12) using the best models of each of the classifiers' types (RLDA and LRM). The performance of these best models of generic classifiers was compared between classifiers and between classifier types. The performance of the generic classifiers was compared to the average performance of the equivalent participant-specific classifiers (referred to as Average S-S), when they were run online.

Test #1 Initial performance: Data from study 2 - session 1 were used as a training and testing dataset, and cross-validation was used to estimate the performance of the classifiers. For this test, the Average S-S was calculated as the mean cross-participant classification performance of study 2 – session 1 (which was an offline session) using the participant-specific dataset and RLDA.

Test #2 Initial generalization performance: This test used the classifiers constructed in Test 1, but data from study 2 - session 2 were used as a testing dataset. Model classification was used to estimate the performance of the classifiers. This test was performed over each participant's data separately, then averaged across all participants. In order to investigate the performance of these generic classifiers, a comparison was made with Average S-S. For this test, Average S-S was calculated as the mean cross-participant classification performance of study 2 – session 2 (which was an online session) using the participant-specific dataset and RLDA.

Table 1 The training and testing details of the performed tests

Test	Training dataset	Testing dataset	Estimating classifiers performance	Average S-S	Nature of data
1 Initial performance	Study 2 - session 1	Study 2 - session 1	Cross-validation	Study 2 – session 1	Offline
2 initial generalization performance	Study 2 - session 1	Study 2 - session 2 (same participants-different session)	Model classification	Study 2 – session 2	Online
3 Primary generalization performance	Study 2 - session 1	Study 1 (completely new participants)	Model classification	Study 1	Offline
4 Initial performance with larger dataset	Study 2	Study 2	Cross-validation	Study 2	Online + offline
5 Initial generalization performance with larger dataset	Study 2	Study 1 (completely new participants)	Model classification	Study 1	Offline
6 Final generalization performance	Study 2 + study 1	Study 2 + study 1	Model classification	Study 2 + study 1	Online + offline

Test #3 Primary generalization performance: Same as Test #2, but data from study 1 were used as a testing dataset, so the datasets were chosen in a way that the testing datasets come from completely different participants. The Average S-S was calculated as the mean cross-participant classification performance of study 1, calculated offline using the participant-specific dataset and RLDA.

Test #4 Initial performance with larger dataset: Same as Test #1, but data from study 2 - sessions 1 to 3 were used as both training and testing datasets. The Average S-S was calculated as the mean cross-participant classification performance of study 2 – sessions 1 to 3, using the participant-specific dataset and RLDA.

Test #5 Initial generalization performance with larger dataset: Same as Test #3, but data from study 2 – sessions 1 to 3 were used as a training dataset. The Average S-S was calculated as the mean cross-participant classification performance of study

1, calculated offline when using the participant-specific dataset-RLDA combinations.

Test #6 Final generalization performance: Data from study 2 - sessions 1 to 3 and data from study 1 were used as both the training and testing datasets. The Average S-S was calculated as the mean cross-participant classification. These six tests are summarized in Table 1.

G. Statistical analysis

Two-way repeated measures ANOVA was used. Pairwise comparisons were performed using Tukey HSD post-hoc test. In all cases, the threshold for statistical significance was set to 0.05. Statistical significance is indicated in the figures by: * for $p < 0.05$, ** for $p < 0.01$ and *** for $p < 0.001$.

III. RESULTS

Test #1 Initial performance

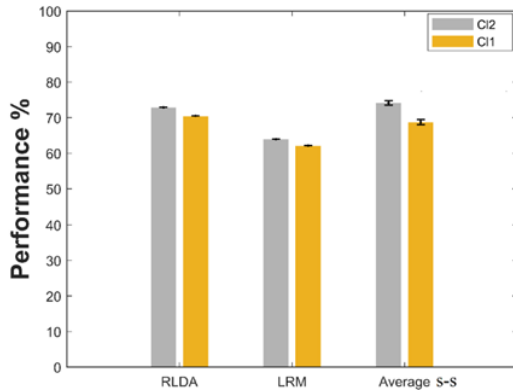


Figure 2: Results of Test #1. Data from study 2 - session 1 were used as a training and testing dataset

Results of test #1 show that when trained and tested on the same lower-limb MI data, the performance of the RLDA classification models reached 70.5% for C11 and 72.9% for C12 (Figure 2). This performance was around 9% higher than the

performance of the LRM classification models (averaged over the two classifiers), which reached 62.1% for C11 and 64.0% for C12. However, RLDA performance was almost equal to the performance of the cross-participant Average S-S.

Test #2 Initial generalization performance

Test #2 shows that when tested on the same participants but on a different session, the cross-participants average performance of the RLDA classification models reached 81.2% for C11 and 85.7% for C12 (Figure 3). The cross-participants average performance of the LRM classification models reached 76.1% for C11 and 71.4% for C12. The Average S-S was 78.9% for C11 and 87.2% for C12.

Repeated measures ANOVA over ‘Algorithm’ and ‘Type’ for performance, yielded a statistically significant intra-participant main effect of “Algorithm” ($F=12.5$, $p < 0.05$) and a significant effect of “Type” ($F=21$, $p < 0.05$). The effect of “Algorithm” x “Type” interaction was not significant. The results of pairwise comparisons over algorithms revealed that the performance of RLDA was significantly better than LRM for C11 (diff = 76.1, $p < 0.05$) and C12 (diff = 14.2, $p < 0.001$). There was no significant difference of algorithm performance between RLDA and Average S-S.

In pairwise comparisons between classifiers within the same algorithm, a significant difference was obtained between C11 and C12 in LRM (diff = 4.6, $p < 0.05$) and in Average S-S (diff = 8.2, $p < 0.01$).

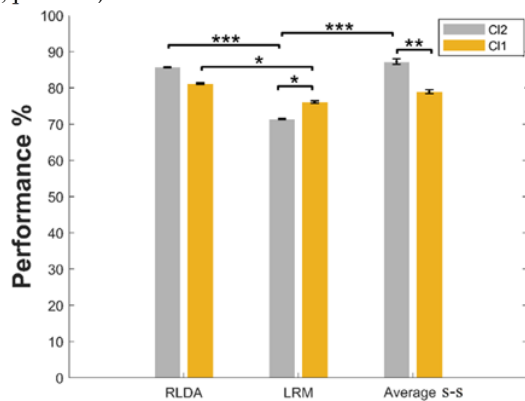


Figure 3: Results of Test #2. Data from study 2 - session 1 were used as a training dataset, but data from study 2 - session 2 were used as a testing dataset.

Test #3 Primary generalization performance

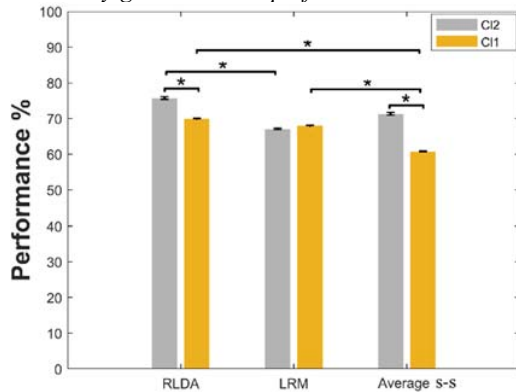


Figure 4: Results of Test #3. Data from study 2 - session 1 were used as a training dataset, and data from study 1 were used as a testing dataset.

The results of Test #3 show that when tested on different participants, the cross-participant average performance of the RLDA classification models reached 69.9% for C11 and 75.7% for C12 (Figure 4). The cross-participant average performance of the LRM classification models reached 68.0% for C11 and 67.1% for C12. The Average S-S was 60.8% for C11 and 71.3% for C12.

Repeated measures ANOVA over ‘Algorithm’ and ‘Type’ for performance, yielded a statistically significant intra-participant main effect of “Algorithm” ($F=11.3$, $p < 0.05$) and a significant effect of “Type” ($F=19$, $p < 0.05$). The effect of the “Algorithm” x “Type” interaction was not significant. The results of pairwise comparisons over algorithms revealed that the performance of RLDA was significantly better than LRM for C12 only (diff = 8.6, $p < 0.05$). There was a significant difference of algorithm performance between RLDA and Average S-S only for C11 (diff = 9.1, $p < 0.05$).

In pairwise comparisons between classifiers within the same algorithm, a significant difference was obtained between C11

and C12 in RLDA (diff = 5.7, $p < 0.05$) and in Average S-S (diff = 10.4, $p < 0.05$).

Test #4 Initial performance with larger dataset

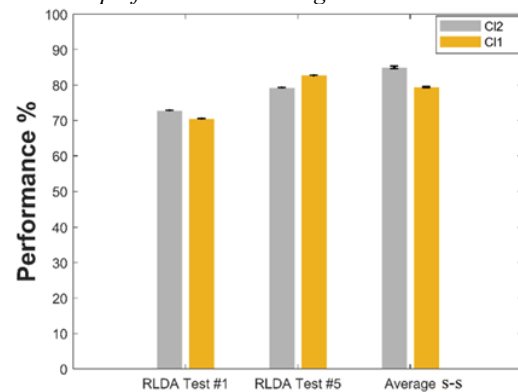


Figure 5: Results of Test #4. Data from study 2 - sessions 1 to 3 were used as both a training and testing dataset. Cross-validation was used to estimate the classifiers’ performance.

The results show that when trained and tested on the same lower-limb MI data, using the data of 3 sessions from the same participants, the performance of the RLDA classification models reached 82.7% for C11 and 79.3% for C12. This performance was a bit lower (3%) than the performance of the cross-participants Average S-S, which reached 79.4% for C11 and 85% for C12.

Test #5 Initial generalization with larger dataset

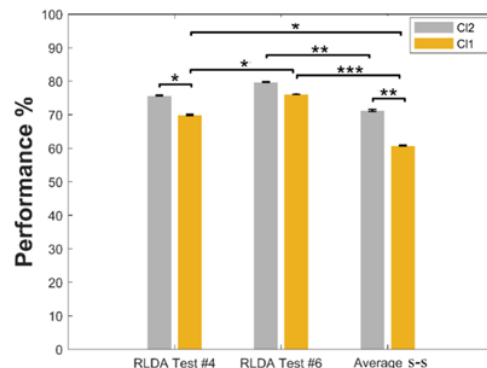


Figure 6: Results of Test #5. Data from study 2 - session 1 to 3 were used as a training dataset and data from study 1 were used as a testing dataset.

The results show that when tested on completely different participants and using a larger dataset to train the classifiers, the cross-participant average performance of the RLDA classification models reached 79.3% for C11 and 82.7% for C12 (Figure 6). The cross-participants Average S-S was 60.8% for C11 and 71.3% for C12.

Repeated measures ANOVA over ‘Algorithm’ and ‘Type’ for performance, yielded a statistically significant intra-participant main effect of “Algorithm” ($F=29.1$, $p < 0.01$) and a significant effect of “Type” ($F=35.1$, $p < 0.05$). The effect of “Algorithm” x “Type” interaction was not significant. The results of pairwise

comparisons over algorithms revealed that the performance of RLDA was significantly better than Average S-S over C11 (diff = 9.3, $p < 0.001$) and C12 (diff = 7., $p < 0.01$). There was a significant difference of algorithm performance between RLDA Test #3 and RLDA Test #5 only for C11 (diff = 9.3, $p < 0.05$). In pairwise comparisons between classifiers within the same algorithm, a significant difference was obtained between C11 and C12 only for Average S-S (diff = 10.4, $p < 0.01$).

Test #6 Final generalization performance

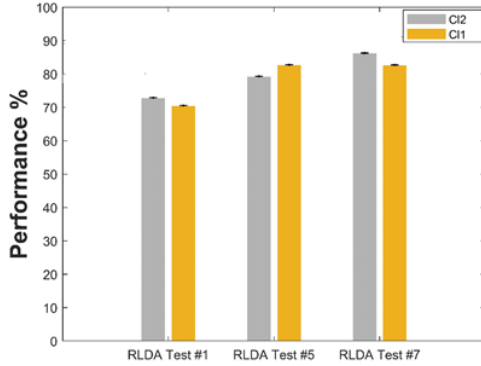


Figure 7: Results of Test #6. Data from study 2 - sessions 1 to 3 and data from study 1 were used as both a training and testing dataset. Cross-validation was used to estimate the classifiers' performance.

The results show that the performance of the RLDA classification models reached 82.7% for C11 and 86.2 % for C12 (Figure 7). This performance was a bit higher (3%) than the performance of the classifiers when trained on data from only study 1.

IV. DISCUSSION

This study focused on the possibility of using two large datasets in order to construct a novel generic classifier that can classify between MI of left steps, rights steps and idle (no movement), and to construct another novel generic classifier that can classify between forward walking and idle (no movement).

Test #1 Initial performance

The results of this test show that when trained and tested on the same lower-limb MI data, the cross-validation performance of the RLDA classification models reached around 71% averaged over the two classifiers, which was not different from when using the participant-specific classifiers.

As stated by Kübler et al. [24], in the field of BCI, the offline classification accuracy for a MI-BCI is considered to be good if it reaches more than 70%. This, however, is in the context where this classifier is used only as a participant-specific classifier, and not as a generic classifier. For a generic classifier, the offline performance that was obtained in other studies was $> 80\%$ [13, 25]. A direct comparison to our results cannot be made since, to our knowledge, this is the first study that uses generic classifiers for lower-limb MI. It is also the first that specifically uses RLDA classification for discrimination of right steps, left steps and no movement. The results of this test

mean that the classification models obtained would not be suitable as generic classifiers.

Taken alone, this result would suggest that the reasons behind the low performance obtained in this test could either be due to the small amount of data (only 20 participants), the sparsity of the data (20 participants with only one session), the complexity of the data (gait data) or the offline nature of the data (no self-regulation of the brain activity to produce better control commands due to the absence of training). The results of test #2 support this statement.

Test #2: Initial generalization performance

The performance increased from approximately 71% to approximately 83% (averaged over the two classifiers).

This unexpected behavior of the classifiers between the use of data in test #1 and the use of the data in test #2 could be explained by the fact that, although the testing data were new data, they were from the same participants. Moreover, they were not only from another session, but also from an online session. In this session, participants received visual feedback, so they could improve the MI signals (datasets) they were trying to produce. Session 1, on the other hand, lacked feedback and participants didn't know how well they were doing in terms of MI performance. Therefore, the main reasons behind the low performance obtained in test #1 could be the small amount of data and the nature of this data.

Test #3 Primary generalization performance

This is the first test in this study where completely new data from new participants were used as testing data. The performance of the generic RLDA classification models were significantly better than when using the participant-specific classifiers but the increase in the performance was only 7%, which is considered poor when compared to other studies [11, 13]. Thus, neither the 72% of performance (averaged over the two classifiers), nor the 7% increase are considered to be good for a generic classifier. In this test, the testing data come from a session which lacked feedback, and participants didn't know how they were doing in terms of MI performance. They therefore had no way to improve their MI and BCI control skills. This problem extends to the training data as well. This could explain the low performance results obtained in this test. Test #2 (Initial generalization performance) used better datasets in an attempt to increase performance.

Test #4 Initial performance with larger dataset

In this test, the use of a larger dataset to train the classifiers led to a performance increase of 10% (to 81%), compared to the use of a dataset from only one session.

Compared to the results of other studies, such as the study of Vidaurre et al. in 2011 [12], and considering the complexity of the MI tasks being carried out in this study, the performance is satisfactory. Although the performance of these classification models was a bit lower than what was obtained when using the participant-specific classifiers, these new RLDA classification models were tested on completely new data, as lower-limb MI generic classifiers.

Test #6 Initial generalization with larger dataset

The results of this test revealed that the new generic RLDA classification models had significantly increased performance, by 10%, compared to participant-specific classifiers. This increase in performance was consistent with what other studies have found [11, 13].

The performance of 81% of these new RLDA classification models was a little bit lower than what was found by other groups such as Lotte and Guan in 2009 [11] (upper-limbs BCI) and this might be due to the complexity of MI tasks, or the small amount of data used to train the classifiers, which is a main limitation for this test.

In order to construct generic classifiers with a satisfactory performance, a large amount of data is required. For example, Fazli et al. [19] used a large database of EEG recordings from 45 Participants. Also, Arvaneh et al. [13] recruited 80 participants and used their data to train the generic classifier. These numbers of participants are more than double and quadruple the number of participants recruited for study 2. One can assume that if there were more participants in study 2, and more sessions from study 2, the performance would have increased further.

Test #7 Final generalization performance

The results revealed that including the datasets of all participants significantly increased the performance, but the increase was modest and for only one classifier (CI2). As with previous tests, this may be due the fact that the added dataset was from a session which lacked feedback. It is probable that if there were feedback sessions in study 1, the performance would have increased further.

The datasets that were fed to the classifiers, from studies 1 and 2, come from all the participants of those studies, even those with lower performance. This may have contributed to a decrease in performance, not only in this test but in all previous tests as well. It is our belief the training datasets should only include the data of the participants who had the best performance, which would consequently require more participants.

V. CONCLUSION

This study reports on the first successful development of two novel generic classifiers that can classify between 4 MI commands: left step versus right step versus no movement and forward walking versus no movement. The generic classifiers were constructed and tested using a dataset of 40 participants performing MI tasks in an immersive virtual environment, from two different studies and a total of 4 sessions. The generic classifiers reached a performance of 87%, similar to the same classifiers using participant-specific data. These results indicate that these generic classifiers will be able to eliminate the calibration phase of a lower-limbs MI-BCI, and thus shorten the time required to learn to control this type of BCIs. Thus, for future work, these classifiers can be implemented on-line with an adaptive RLDA, in order to be used later in a BCI to control the gait of an avatar.

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